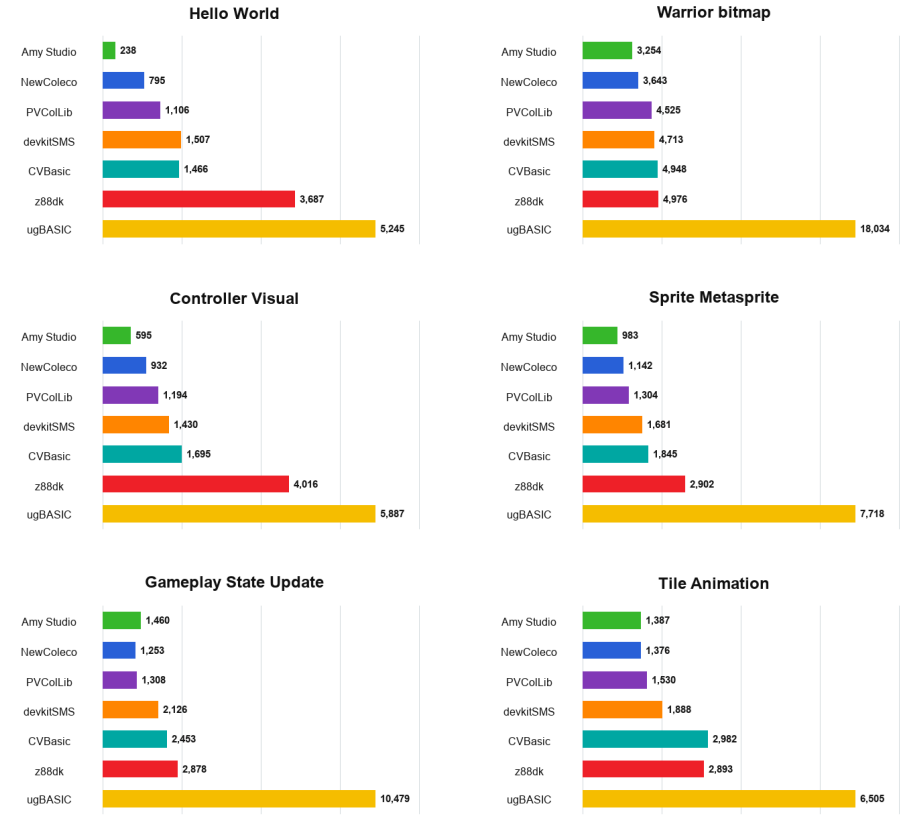
Amy Studio ColecoVision Toolchain Comparison

Measured results, capabilities, compression evidence, and development priorities

Real occupied size, excluding cartridge padding

sizes in bytes (lower is better); columns follow the six-sample total ranking



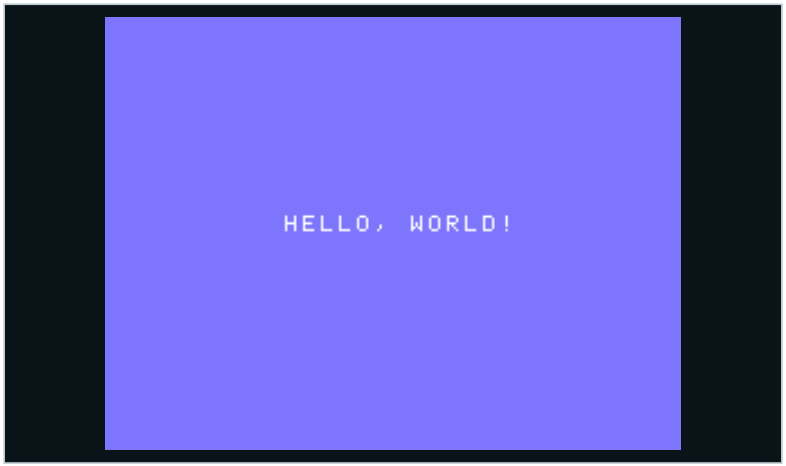
Occupied ROM size excludes cartridge padding. Lower is smaller, not automatically faster or better.

Amy Studio source and GearColeco results

Listings are complete except for bulky sprite patterns and the state-update engine body; both complete sources are in the evidence archive. Screenshots come from GearColeco 1.6.8 running each measured ROM for 180 NTSC frames.

Hello World

```
text screen
backdrop sky blue
print centered at 11, "HELLO, WORLD!"
screen on
```



GearColeco screenshot

GearColeco output after 180 NTSC frames

Warrior Graphics II Bitmap

```
picture WarriorPicture:
  pattern from "@project/warrior.pattern.zx0" codec zx0
  color from "@project/warrior.color.zx0" codec zx0
end picture

show picture WarriorPicture
```

GearColeco screenshot

Exact GearColeco output; all seven tools reproduce the same target image

Visual Controller Input

```
text screen
backdrop dark blue
print centered at 8, "CONTROLLER TEST"
print centered at 11, "MOVE OR PRESS A BUTTON"
print centered at 14, "KEYPAD IS PURPLE"
screen on

InputLoop:
  wait
  if keypad(1) <> 255 then
    backdrop magenta
  elseif joypad(1).fire then
    backdrop light red
  elseif joypad(1).up then
    backdrop light green
  elseif joypad(1).down then
    backdrop dark green
  elseif joypad(1).left then
    backdrop light yellow
  elseif joypad(1).right then
    backdrop cyan
  else
    backdrop dark blue
  end if
  goto InputLoop
```

GearColeco screenshot

GearColeco output at the neutral controller state

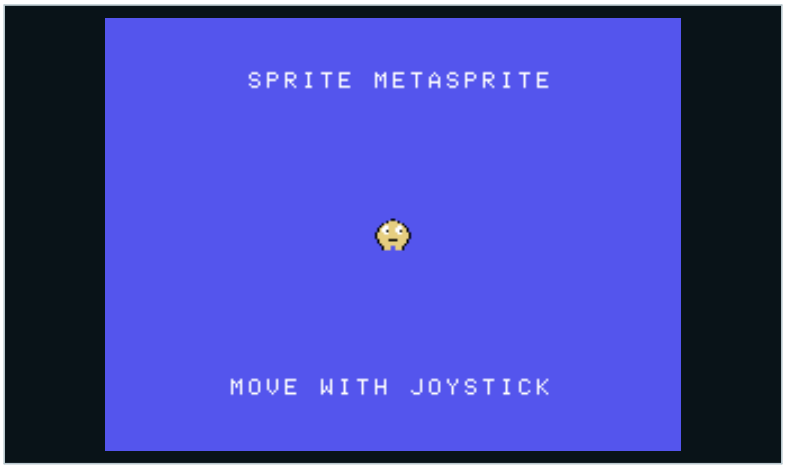
Animated Three-Color Metasprite

```
u8 PlayerX = 120
u8 PlayerY = 88
u8 AnimationTick = 0
u8 AnimationFrame = 0

data PlayerMeta metasprite layers 3
  frame 0,15, 4,11, 8,1
  frame 12,15, 16,11, 20,1
end data

' Six 16x16 pattern blocks are included in the evidence archive.
sprites simple
set sprite count 3

MainLoop:
  wait
  if joypad(1).left and PlayerX > 1 then PlayerX -= 1
  if joypad(1).right and PlayerX < 239 then PlayerX += 1
  if joypad(1).up and PlayerY > 1 then PlayerY -= 1
  if joypad(1).down and PlayerY < 175 then PlayerY += 1
  AnimationTick += 1
  if AnimationTick = 8 then
    AnimationTick = 0
    AnimationFrame = 1 - AnimationFrame
  end if
  set metasprite PlayerMeta frame AnimationFrame to PlayerY,PlayerX using sprite 0
  update sprites
  goto MainLoop
```



GearColeco screenshot

A metasprite is one visual actor assembled here from three overlapping hardware sprites

Gameplay State Update

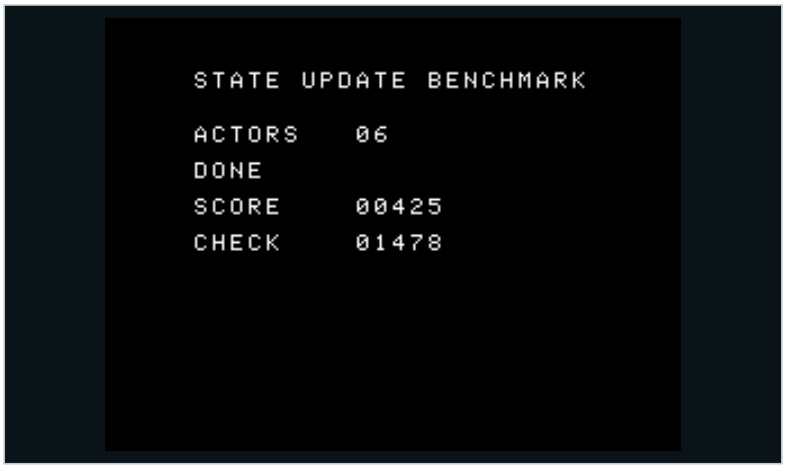
```
record ActorState:
  u8 X
  u8 Y
  u8 Direction
  u8 Timer
  u8 Mode
  u8 Hits
end record

state machine WorldBehavior:
  Running calls UpdateActors
  Bonus calls UpdateBonus
end state machine

ActorState Actors[6]
u16 Score = 0
u16 Checksum = 0

for Tick = 0 to 47
  dispatch WorldMode using WorldBehavior
next

sub UpdateActors:
  for each Actor, ActorIndex in Actors
    ' Movement, timers, modes, collision and score updates.
  next Actor
  if Tick = 23 then WorldMode = WorldBehavior.Bonus
  return
```



GearColeco screenshot

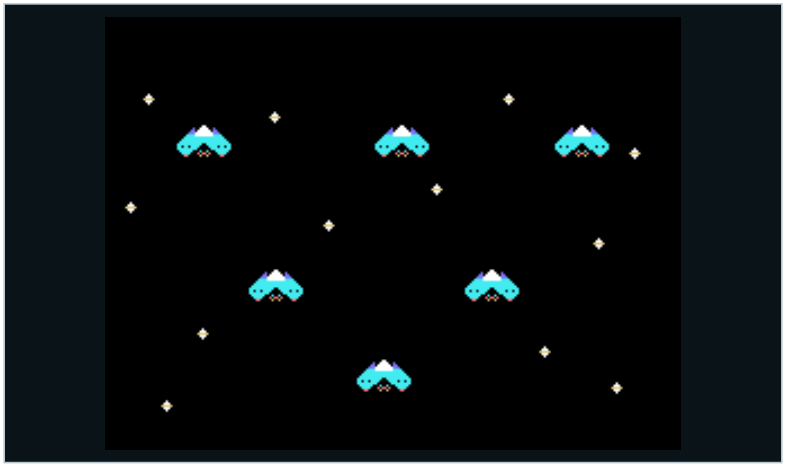
Experimental: 1,460 occupied bytes; exact oracle: state 1, collisions 13, score 425, checksum 1478

Graphics II Tile Animation

```
const StarTile = $80
const ShipFirstTile = $90
u8 StarFrame = 0
u8 ShipFrame = 0

tile screen
backdrop black
nmi off
' Install shared star and two 3x2 ship frames in all Graphics II thirds.
copy StarPattern0 count 8 to vram.pattern + StarTile * 8
copy ShipPatterns count 96 to vram.pattern + ShipFirstTile * 8
copy ShipColors count 96 to vram.color + ShipFirstTile * 8
nmi on

AnimationLoop:
  wait
  Tick += 1
  if Tick & 3 = 0 then
    ' Replace one shared pattern: every star animates together.
    StarFrame = (StarFrame + 1) % 3
    copy StarPatterns[StarFrame * 8] count 8 to vram.pattern + StarTile * 8
  end if
  if Tick & 7 = 0 then
    ' Replace six 3x2 Name Table frames without moving the ships.
    ShipFrame = 1 - ShipFrame
    for Ship = 0 to 5
      put ShipTiles[ShipFrame * 6] frame size 3,2 at ShipX[Ship],ShipY[Ship]
    next
  end if
goto AnimationLoop
```



GearColeco screenshot

Experimental: 1,387 occupied bytes; continuous shared-pattern and six-ship Name Table animation verified

Graphics II compression corpus

All 42 unique 12,288-byte pictures passed exact round-trip checks with every codec. The gallery shows the actual TMS9918 output, the three smallest payloads, and Pletter's rank when it falls outside the top three. Exomizer also needs a 226-byte ROM decoder and 156 bytes of CPU RAM. Amy Studio reserves that workspace at a 256-byte boundary; its generated address varies and must not overlap other RAM.

Rank	Codec	Payload total	Decoder	First-use places (pictures)				
				1st	2nd	3rd	4th	5th
1	Exomizer 2	161,374	226	21	12	2	3	1
2	ZX0	166,556	133	21	18	1	2	0
3	DAN2	166,923	212	0	0	7	13	10
4	DAN3	167,186	205	0	0	16	5	11
5	DAN1	167,240	205	0	3	4	9	16
6	MegaLZ	171,016	162	0	0	1	1	1
7	aPLib Compact	171,484	244	0	0	0	0	0
8	Pletter	171,751	212	0	0	0	0	0
9	ZX7	171,788	136	0	0	2	9	3
10	BitBuster	172,941	166	0	0	0	0	0
11	ZX1	176,562	127	0	4	5	0	0
12	ZX2	177,582	115	0	5	4	0	0
13	LZF	193,412	117	0	0	0	0	0
14	Nibble	214,555	115	0	0	0	0	0
15	MDK-RLE	253,859	46	0	0	0	0	0

Corpus gallery 1-6

Cake
1. Exomizer 2: 2,935 bytes
2. ZX0: 2,958 bytes
3. DAN2: 2,999 bytes
... 9. Pletter: 3,108 bytes

Commando
1. Exomizer 2: 5,206 bytes
2. ZX0: 5,310 bytes
3. DAN2: 5,312 bytes
... 8. Pletter: 5,508 bytes

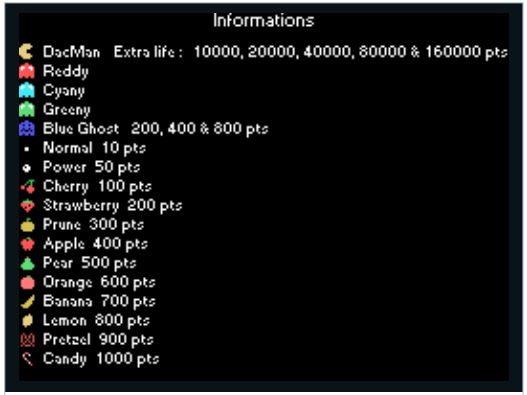
Warrior
1. Exomizer 2: 2,821 bytes
2. ZX0: 2,849 bytes
3. DAN3: 2,891 bytes
... 10. Pletter: 2,988 bytes

Barbarian
1. Exomizer 2: 4,165 bytes
2. ZX0: 4,229 bytes
3. DAN3: 4,286 bytes
... 5. Pletter: 4,339 bytes

421 title
1. ZX0: 1,055 bytes
2. Exomizer 2: 1,060 bytes
3. DAN2: 1,067 bytes
... 8. Pletter: 1,110 bytes

Dacman title
1. Exomizer 2: 967 bytes
2. DAN1: 969 bytes
3. DAN2: 972 bytes
... 9. Pletter: 1,035 bytes

Corpus gallery 7-12

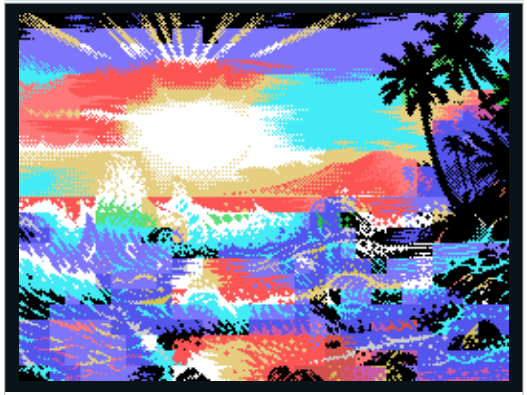


Dacman info
1. Exomizer 2: 1,248 bytes
2. ZX0: 1,260 bytes
3. DAN2: 1,273 bytes
... 10. Pletter: 1,361 bytes

Dacman 2 title
1. Exomizer 2: 1,142 bytes
2. DAN2: 1,168 bytes
3. DAN1: 1,171 bytes
... 8. Pletter: 1,231 bytes

Chateau title
1. Exomizer 2: 1,613 bytes
2. ZX0: 1,636 bytes
3. DAN1: 1,651 bytes
... 10. Pletter: 1,730 bytes

Almighty God - Aliquis Acuario Ingens
1. Exomizer 2: 7,099 bytes
2. ZX0: 7,234 bytes
3. DAN3: 7,271 bytes
... 7. Pletter: 7,447 bytes

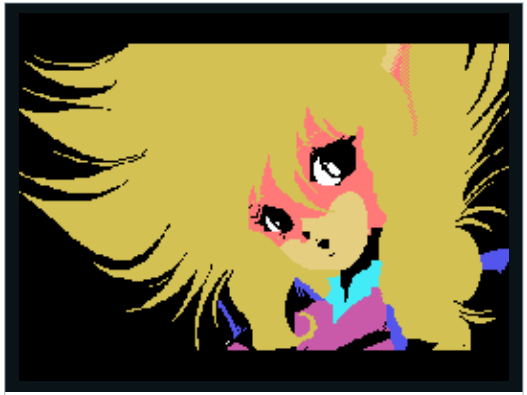


Aloha (ArtField 2010, 1)
1. Exomizer 2: 6,754 bytes
2. DAN1: 6,989 bytes
3. DAN2: 6,991 bytes
... 9. Pletter: 7,199 bytes

Aorante - Lady Saturn
1. Exomizer 2: 3,497 bytes
2. DAN3: 3,653 bytes
3. DAN2: 3,657 bytes
... 8. Pletter: 3,810 bytes

Corpus gallery 13-18

bejeweled title public.pp
1. Exomizer 2: 1,519 bytes
2. ZX0: 1,564 bytes
3. DAN1: 1,594 bytes
... 9. Pletter: 1,661 bytes

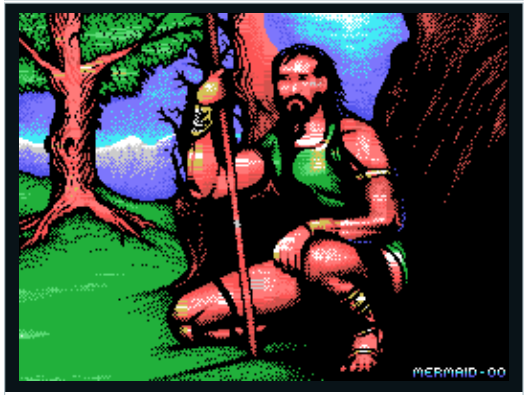


benchmark-anime-portrait
1. Exomizer 2: 2,098 bytes
2. DAN2: 2,157 bytes
3. DAN3: 2,162 bytes
... 8. Pletter: 2,265 bytes

benchmark-band-logo
1. Exomizer 2: 2,155 bytes
2. ZX0: 2,172 bytes
3. DAN2: 2,183 bytes
... 6. Pletter: 2,243 bytes

benchmark-bat-character
1. Exomizer 2: 945 bytes
2. ZX0: 980 bytes
3. DAN1: 987 bytes
... 8. Pletter: 1,051 bytes

benchmark-blue-portrait
1. Exomizer 2: 5,902 bytes
2. ZX0: 6,073 bytes
3. DAN2: 6,097 bytes
... 7. Pletter: 6,224 bytes



benchmark-fantasy-camp
1. Exomizer 2: 5,996 bytes
2. DAN2: 6,138 bytes
3. DAN1: 6,147 bytes
... 8. Pletter: 6,317 bytes

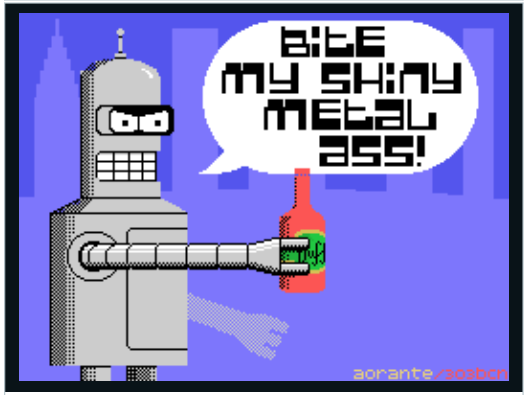
Corpus gallery 19-24

benchmark-feliz-navidad
1. Exomizer 2: 3,998 bytes
2. DAN3: 4,149 bytes
3. ZX0: 4,154 bytes
... 9. Pletter: 4,300 bytes

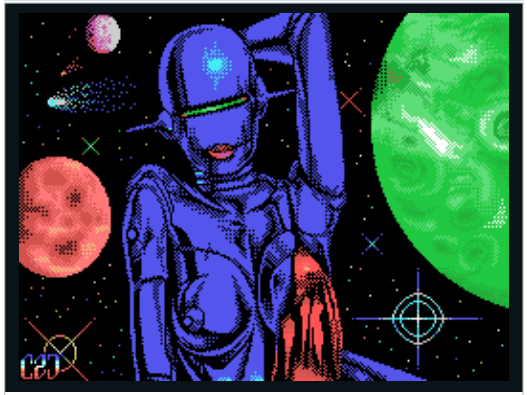
benchmark-laser-squad
1. Exomizer 2: 7,357 bytes
2. DAN2: 7,488 bytes
3. DAN1: 7,494 bytes
... 8. Pletter: 7,774 bytes

benchmark-mermaid-portrait
1. Exomizer 2: 5,848 bytes
2. ZX0: 6,050 bytes
3. DAN2: 6,070 bytes
... 7. Pletter: 6,196 bytes

benchmark-reservoir-dogs
1. Exomizer 2: 2,208 bytes
2. ZX0: 2,315 bytes
3. DAN3: 2,316 bytes
... 8. Pletter: 2,414 bytes



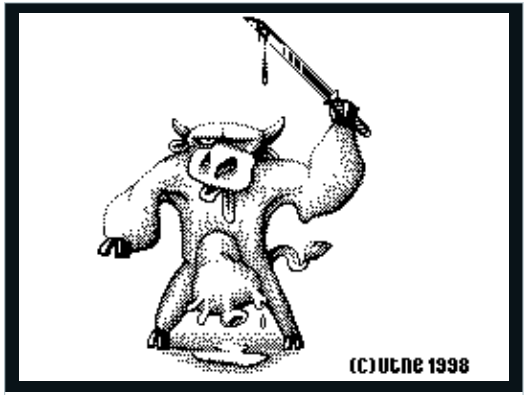
benchmark-robot-quote
1. Exomizer 2: 1,990 bytes
2. DAN3: 2,076 bytes
3. DAN2: 2,097 bytes
... 9. Pletter: 2,217 bytes



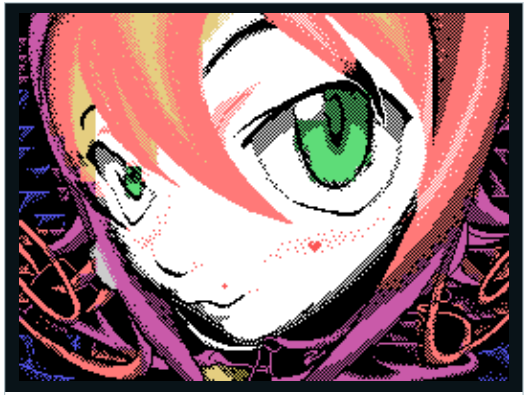
benchmark-space-soldier
1. Exomizer 2: 5,522 bytes
2. DAN2: 5,649 bytes
3. DAN1: 5,661 bytes
... 9. Pletter: 5,886 bytes

Corpus gallery 25-30

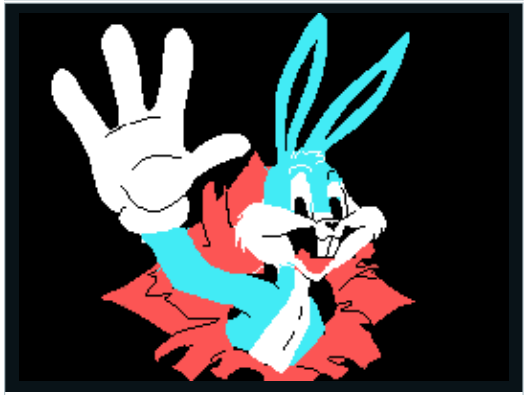
benchmark-turban-character
1. Exomizer 2: 7,629 bytes
2. DAN1: 7,908 bytes
3. DAN2: 7,911 bytes
... 7. Pletter: 8,170 bytes



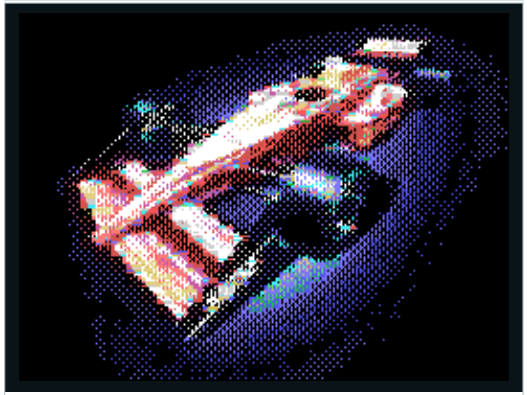
benchmark-wizard-sketch
1. Exomizer 2: 2,125 bytes
2. DAN3: 2,171 bytes
3. DAN2: 2,180 bytes
... 9. Pletter: 2,273 bytes



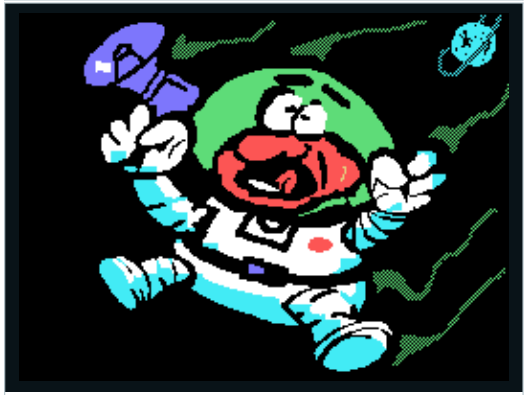
Blair (Chaos Constructions 2009, 6)
1. Exomizer 2: 5,140 bytes
2. ZX0: 5,372 bytes
3. DAN3: 5,374 bytes
... 8. Pletter: 5,491 bytes



Bugs (Demobit 1995, 4)
1. Exomizer 2: 2,032 bytes
2. DAN3: 2,091 bytes
3. ZX0: 2,107 bytes
... 9. Pletter: 2,191 bytes

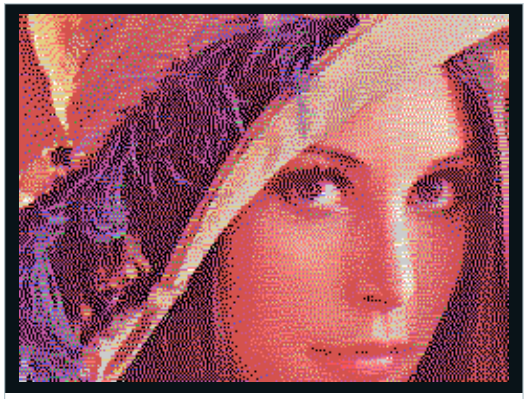


f1spp
1. Exomizer 2: 4,169 bytes
2. DAN3: 4,277 bytes
3. DAN2: 4,278 bytes
... 9. Pletter: 4,462 bytes



Kozmonaut (Forever 7, 7)
1. Exomizer 2: 2,694 bytes
2. DAN3: 2,780 bytes
3. ZX0: 2,792 bytes
... 9. Pletter: 2,892 bytes

Corpus gallery 31-36

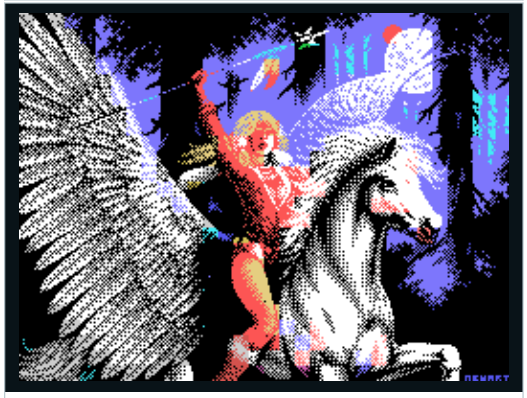


lenna
1. Exomizer 2: 9,071 bytes
2. ZX0: 9,400 bytes
3. DAN2: 9,425 bytes
... 8. Pletter: 9,556 bytes

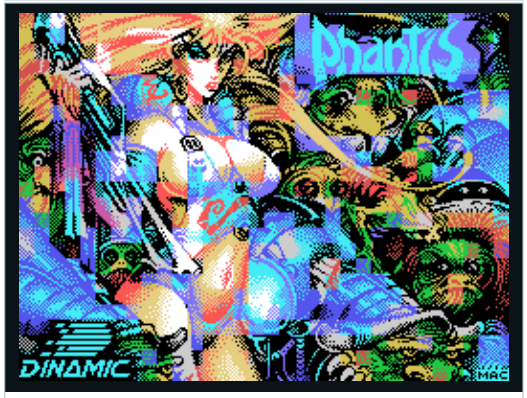
maze maniac.pp
1. Exomizer 2: 2,625 bytes
2. DAN3: 2,709 bytes
3. ZX0: 2,713 bytes
... 7. Pletter: 2,817 bytes

mona lisa
1. Exomizer 2: 9,208 bytes
2. ZX0: 9,426 bytes
3. DAN3: 9,490 bytes
... 5. Pletter: 9,640 bytes

NewColeco ROM File Edition
1. ZX0: 901 bytes
2. Exomizer 2: 904 bytes
3. DAN2: 909 bytes
... 9. Pletter: 960 bytes

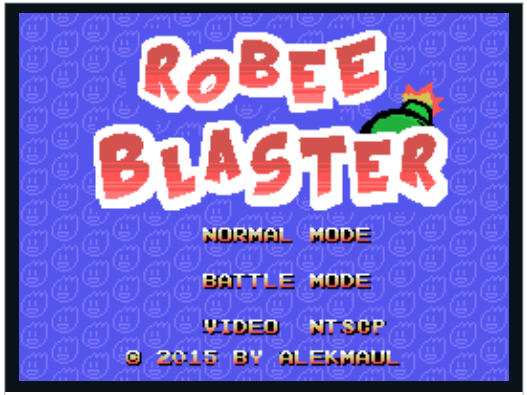


Pegasus (Millennium 1903, 1)
1. Exomizer 2: 6,168 bytes
2. ZX0: 6,405 bytes
3. DAN1: 6,415 bytes
... 8. Pletter: 6,544 bytes



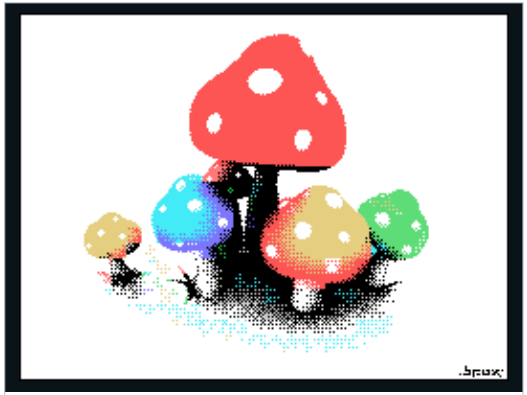
Phantis (3BM OpenAir 2014, 1)
1. Exomizer 2: 8,109 bytes
2. DAN1: 8,387 bytes
3. DAN2: 8,388 bytes
... 8. Pletter: 8,599 bytes

Corpus gallery 37-42



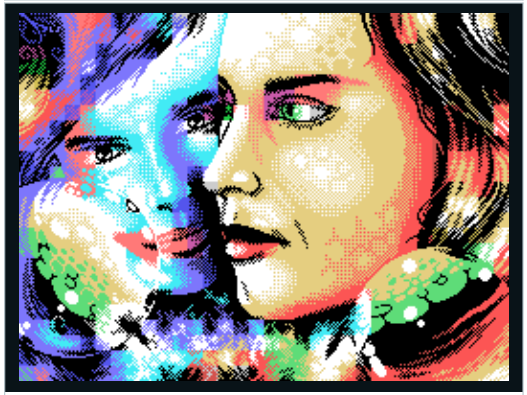
Robee Blaster title
1. Exomizer 2: 2,024 bytes
2. ZX0: 2,082 bytes
3. DAN2: 2,104 bytes
... 8. Pletter: 2,138 bytes

smurf challenge
1. Exomizer 2: 1,227 bytes
2. DAN1: 1,264 bytes
3. DAN2: 1,269 bytes
... 8. Pletter: 1,342 bytes



Stainly Rainbow (ArtField 2006, 3)
1. Exomizer 2: 2,170 bytes
2. ZX0: 2,248 bytes
3. DAN2: 2,253 bytes
... 8. Pletter: 2,358 bytes

Vranov (Chaos Constructions 2001, 7)
1. Exomizer 2: 3,396 bytes
2. ZX0: 3,475 bytes
3. DAN3: 3,494 bytes
... 8. Pletter: 3,599 bytes



Xzema (Chaos Constructions 2001, 2)
1. Exomizer 2: 6,544 bytes
2. DAN1: 6,798 bytes
3. DAN2: 6,815 bytes
... 9. Pletter: 7,079 bytes

Znachok (Enlight 1996, 15)
1. Exomizer 2: 2,094 bytes
2. DAN3: 2,134 bytes
3. ZX0: 2,151 bytes
... 6. Pletter: 2,226 bytes

Date: 2026-09-01

Method

This comparison separates four kinds of evidence:

- **Runtime verified**: compiled for ColecoVision and checked in an emulator or self-test.
- **Build verified**: the ColecoVision source, header, library, or generated assembly was inspected.
- **Documented**: claimed by the tool's official documentation, but not yet measured here.
- **Unverified**: available somewhere in the wider tool, but not proven for ColecoVision.

Amy Studio combines a language and IDE; the others are mainly compilers or C kits. IDE-only features are listed separately from language capabilities.

The seven solutions

Solution	Primary approach	ColecoVision position
Amy Studio	ColecoVision-first Amy language and browser IDE	Original 1 KB RAM, BIOS-aware, no extra hardware by design
CVBasic	Portable integer BASIC cross-compiler	Mature native target; optional MegaCart and SGM support
z88dk `+coleco`	General Z80 C toolchain	C compiler, assembler, linker, libraries, and app packaging
ugBASIC	Portable retro BASIC compiler	Declared target with common high-level multimedia vocabulary
devkitSMS + `SGlib_CV`	SDCC C plus compact game libraries	ColecoVision adaptation of the SG/SMS development workflow
PVColLib	ColecoVision-focused SDCC C library and devkit	Native VDP, controller, sprite, sound, music, and compression APIs
NewColeco	Historical SDCC C, `CRTCV`, `CVLIB`, and GETPUT 1.1	Amy's pre-Studio ColecoVision workflow and direct ancestor of current techniques

Reproducible six-sample ROM suite

The suite builds six runnable programs with all seven solutions:

1. a visible Hello World;
2. the Warrior Graphics II bitmap picture;
3. a visual controller monitor;
4. an animated, controller-driven three-color metasprite;
5. a deterministic six-actor gameplay state update;
6. a Graphics II star field and six animated 3x2 tile ships.

A **metasprite** combines hardware sprites into one actor. This test overlaps three 16x16 layers (white, yellow, black), below the four-sprites-per-scanline limit.

The build scripts create 42 ROMs, grouped under `build/competition`. `tools/report-five-tool-sample-sizes.ps1` records occupied sizes; dedicated GearColeco tests verify the bitmap, controller, metasprite, and deterministic state-update oracles.

Maximum native optimization used

Solution	Setting used	Observation
Amy Studio	Experimental	Saved 1 byte on Bitmap and 2 on Controller versus Balanced; Hello was unchanged
CVBasic 0.9.2	Normal compiler; TMSColor `-z -p2` for Bitmap	No stronger compiler optimization switch was exposed; Pletter is used for its bitmap
z88dk	`+coleco -O3`; ZX0 for Bitmap	Coleco-safe direct-to-VRAM port of z88dk's ZX0 core; ZX7, MDKRLE, and raw baselines remain measured
ugBASIC 1.18	Default maximum of 16 peephole passes	Explicit 32- and 64-pass builds produced the same Hello binary
devkitSMS / SDCC 4.5	`--opt-code-size --max-allocs-per-node 100000` on the program and SGlib	Saved 60 bytes on Bitmap; Hello and Controller were unchanged
PVColLib 1.6.0 / bundled SDCC	`--opt-code-size --max-allocs-per-node 20000`	Official build flags; linked only referenced library modules
NewColeco / SDCC 3.8	`--std-c99`; original prebuilt libraries plus historical DAN2	The exact DAN2 bitmap is 626 bytes smaller than its GETPUT MDKRLE baseline

These are the strongest **native settings validated here**. No competitor output passed through Amy's optimizer or MDL. Amy Experimental serves this size test; Balanced remains its default.

Real occupied size, excluding cartridge padding

Sample	Amy Studio	NewColeco	PVColLib	devkitSMS	CVBasic	z88dk	ugBASIC
Hello World	**238**	795	1,106	1,507	1,466	3,687	5,245
Warrior bitmap	**3,254**	3,643	4,525	4,713	4,948	4,976	18,034
Controller Visual	**595**	932	1,194	1,430	1,695	4,016	5,887
Sprite Metasprite	**983**	1,142	1,304	1,681	1,845	2,902	7,718
Gameplay State Update	1,460	**1,253**	1,308	2,126	2,453	2,878	10,479
Tile Animation	1,387	**1,376**	1,530	1,888	2,982	2,893	6,505
Six-sample total	**7,917**	**9,141**	**10,967**	**13,345**	**15,389**	**21,352**	**53,868**

Measured bitmap baselines: z88dk RAW 14,293 bytes, MDKRLE 5,911, ZX7 5,115, and ZX0 4,976; NewColeco GETPUT/MDKRLE 4,269 and DAN2 3,643. All reproduce both VRAM tables and 49,152 pixels. ugBASIC's `COMPRESSED` and `NONE` Warrior builds are identical at 18,034 bytes: MSC1 was discarded, and its RLE image branch is C128-only.

Legacy payloads are MDKRLE 3,687, DAN1 2,903, DAN2 2,897, and DAN3 2,891 bytes. DAN2 stays linked: DAN3 saves six payload bytes before decoder cost, while DAN2 is validated end to end.

`Occupied` includes header, runtime, code, and data, but excludes cartridge padding. Sources are: assembled length (Amy), `ROM_END-$8000` (CVBasic), unpadded binary (z88dk), generated code+data (ugBASIC), Intel HEX span above `$8000` (devkitSMS/PVColLib), and complete linked binary (NewColeco).

Runtime and comparability verdict

Sample	Runtime result	Verdict
Hello World	Seven ROMs complete 180 GearColeco frames	Stable startup
Warrior bitmap	Seven native pipelines render the same 256x192 image	`0 / 49,152` pixels differ
Controller Visual	Six pass injected neutral, keypad, UP, FIRE, and release states	Partial: ugBASIC does not update VDP R7
Sprite Metasprite	Seven pass the same VRAM and sprite-table checks	Exact patterns, layers, and priority
Gameplay State Update	Seven match world state 1, 13 collisions, score 425, checksum 1478	Exact deterministic oracle
Tile Animation	Seven continuously animate shared patterns and six 3x2 Name Table frames	Exact VRAM tables, no stray ship tiles, animation continues after 100 frames

PVColLib and NewColeco use `$F0` transparent-background text so VDP R7 changes remain visible; controller logic and occupied size are unchanged.

Native bitmap paths are Amy/z88dk ZX0, CVBasic Pletter, ugBASIC resources, devkitSMS aPLib, PVColLib RLE, and NewColeco DAN2. All framebuffers are exact despite equivalent internal table encodings. PVColLib Pletter failed VRAM validation and is excluded.

For z88dk, MDKRLE reaches VRAM at frame 93, ZX0 at 132, and ZX7 at 138. ZX0 saves 935 bytes versus MDKRLE and 139 versus ZX7; MDKRLE is faster. A transplanted SMS aPLib path failed and is excluded; native devkitSMS aPLib remains exact with frame interrupts disabled during decompression.

The initial ugBASIC mismatch came from the wrong RGB palette. With its target palette, all four corpus pictures render exactly.

Observations supported by this suite

- Amy is smallest for Hello, Controller, and this exact bitmap ROM (3,254 versus CVBasic's 4,948 bytes).
- Portable C/BASIC runtimes add visible fixed cost, but complete games may scale differently.
- Cartridge length includes packaging; a padded 32 KB file is not necessarily a 32 KB program.
- Smaller results count only after runtime validation; five samples cannot prove a universal winner.
- RAM, worst-frame cycles, build latency, sound, sprite pressure, and gameplay remain separate tests.

Preliminary capability matrix

Legend: **Yes** = verified or explicit target support; **Partial** = manual or narrower support; **Pending** = must still be proven specifically on ColecoVision. ***** marks a benchmark adaptation written here, not a native ColecoVision direct-to-VRAM path supplied by that solution.

Capability	Amy Studio	CVBasic	z88dk	ugBASIC	devkitSMS / SGlib_CV	PVColLib
8/16-bit integers	Yes	Yes	Yes	Documented	Yes, through SDCC	Yes, through SDCC
32-bit integers	Yes, runtime tested	No native BASIC type	Yes	Documented; target cost pending	Yes, through SDCC	Yes, through SDCC
Fixed-point	Yes, 8.8 and 16.16	No	Library/manual	Documented; target proof pending	Manual C/library code	Manual C/library code
Records/structures	Yes	No	Yes	Documented custom types	Yes	Yes
2D arrays	Yes	No native 2D syntax	Yes	Documented	Yes	Yes
RAM overlays	Yes, first-class	Manual	C union/manual layout	Pending	C union/manual layout	C union/manual layout
Functions with values and returns	Yes	Limited BASIC procedures/DEF FN	Yes	Yes	Yes	Yes
Inline Z80 assembly	Yes	Yes	Yes	Documented	Yes	Yes
TMS9918 screen and tile operations	Yes	Yes	Yes	Yes, target proof incomplete	Yes	Yes
Hardware sprite API	Yes	Yes	Low-level/library	Yes, exact target surface pending	Yes	Yes, 32-entry SAT
Four-sprite scanline mitigation	Yes, stable ranges and flicker	Yes, sprite flicker	Manual	Pending	Manual/library-dependent	Yes, NMI sprite swapping
Keypad	Yes	Yes	Library/manual	Pending	Yes/target library	Yes, both ports
Spinner / Roller Controller	Yes	Yes	Not confirmed	Pending	Not confirmed	Yes, both ports
Held, pressed, and released input	Yes	Held; edges are manual	Manual	Manual edges over `JOY`	Yes, computed from NMI snapshots	Manual edges over NMI snapshots
Coleco PSG sound	BIOS tables, Tiny Sound, DSOUND	Sound/music commands	Sound libraries	Sound commands, target proof pending	PSGlib_CV	BIOS-style sound tables and sequenced music
Direct-to-VRAM compression	Fifteen active codecs, including Exomizer 2 and MegaLZ	Pletter	RAM APIs; ZX0, *ZX1*, ZX2, *and ZX7* benchmark VRAM ports	Resource conversion; RAM-oriented compression	ZX7 and aPLib	RLE, Pletter, DAN1/2/3
ROM banking	Intentionally no	Yes	Yes	Pending	SMS workflow has banking; CV support pending	MegaCart tools and examples
Source-level Coleco debugger	Integrated	External emulator	External debugger/emulator	External or IDE-dependent	External debugger/emulator	External debugger/emulator
Rewind, breakpoints, VRAM/RAM inspection	Integrated	External	External	External	External	External
Graphics editors and project assets	Integrated	Separate tools	Separate tools	Conversion-oriented IDE/tools	Separate asset tools	`gfx2col` and separate asset tools
Automated ROM behavior tests	Integrated project pipeline	Not supplied as one system	Can be assembled manually	Not established	Not established	Not established

Detailed language and runtime evidence

Data model

Area	Amy Studio	CVBasic	z88dk	ugBASIC	devkitSMS	PVColLib
Integer widths	Explicit signed/unsigned 8/16/32	8/16, signed or unsigned	C 8/16/32	Backend defines signed/unsigned 8/16/32	SDCC C 8/16/32	SDCC C 8/16/32
Fractional math	`fixed`, `ufixed`, `fixed32`, `fp5`	None	C/library float and fixed libraries	`VT_FLOAT` exists; Coleco cost pending	SDCC float/manual fixed	SDCC float/manual fixed
Decimal scores	Packed BCD 1-12 digits	Manual	Manual/library	Manual	Manual/library	Manual/library
Aggregates	Nested records, record arrays	None	struct/union	Typed arrays and internal resource types; user-structure surface pending	struct/union	struct/union
Multidimensional arrays	Primitive 2D arrays	One-dimensional only	Native C	Typed arrays; dimensional coverage pending	Native C	Native C
Mutually exclusive RAM	First-class overlays/scenes	Manual aliases	union/linker/manual	Banking/resource allocator; no equivalent proven	union/linker/manual	union/linker/manual
Dynamic strings	Deliberately fixed-buffer oriented	Mostly literals/printing	C strings and allocation libraries	First-class strings	C strings; risky in 1 KB RAM	C strings; risky in 1 KB RAM

Amy's data model is the most game-oriented for stock ColecoVision RAM. C remains the most general, but generality does not provide automatic lifetime analysis or a debugger-aware overlay. ugBASIC has a broad internal type system; its actual ColecoVision ROM/RAM cost still requires compilation.

Control, timing, and code organization

Area	Amy Studio	CVBasic	z88dk	ugBASIC	devkitSMS	PVColLib
Procedures/functions	Typed values, `ref`, returns, recursion	Procedures without value parameters; `DEF FN` expression macros	Full C	Procedures/functions documented	Full C	Full C
State machines	Typed zero-RAM dispatch	Manual `SELECT`/labels	Manual enum/switch/table	Manual control constructs	Manual enum/switch/table	Manual enum/switch/table
VBlank	`wait`, frame hooks, timers	`WAIT`, `ON FRAME`	HALT/NMI hooks/manual	`WAIT VBL`, `EVERY`, timers	`SG_waitForVBlank`, frame handler	`vdp_waitvblank`, user `nmi` hook
Parallel animation	Not yet first-class	Manual	Manual/interrupt	`ANIMATION`, `ANIMATE`, `MOVE`, paths and multitasking source present	Manual frame service	Manual NMI/frame service
Inline assembly	Amy ASM bridge	`ASM`, `CALL`, `USR`	Native inline/external ASM	Supported	SDCC inline/external ASM	SDCC inline/external ASM

ugBASIC is the strongest source of ideas for a future optional Amy animation service. It also shows the risk: animations create state variables, thread handles, paths, delays, signals, and background-preservation storage. Amy should not copy that surface until exact RAM and cycle costs are visible and the entire service disappears when unused.

TMS9918 graphics and sprites

Area	Amy Studio	CVBasic	z88dk	ugBASIC	SGlib_CV	PVColLib
Text/tile/bitmap modes	Coleco-specific commands and editors	Modes and direct VRAM commands	Generic TMS9918/graphics libraries	TMS9918 backend and resource conversion	Tile, bitmap, VDP and VRAM functions	Text, bitmap, VDP and VRAM functions
Pixel/line/circle	Yes, programmer controls NMI-safe batching	Yes	Generic graphics	TMS9918 drawing backend	Pixel/bitmap helpers	Low-level VRAM/bitmap support

Area	Amy Studio	CVBasic	z88dk	ugBASIC	SGlib_CV	PVColLib
Sprites	Fields, clipping helpers, movement, editor	Define/place sprites	Low-level/library	Sprite conversion and TMS9918 operations	SAT builder, clipping, finalize/copy	32-entry SAT, update and fast-update APIs
Collision	Hitboxes, tile groups, hardware status	VDP status/manual	Hardware/manual	Hardware collision/hit backend	VDP status/manual	`spr_collide` hitboxes and VDP status
Scanline overflow	Explicit flicker and stable priority ranges	Sprite flicker	Manual	No verified policy	Manual ordering	`spr_getentry` swaps sprites each NMI
Asset authoring	Integrated bitmap/tile/sprite/frame editors	Separate tools	Separate tools	Automatic image conversion	External asset tools	`gfx2col` and external tools

Amy's lead is the complete authoring/debugging workflow and explicit TMS9918 policy. ugBASIC's automatic modern-image conversion is substantial, while SGlib_CV offers a compact C SAT workflow.

Controllers

Capability	Amy Studio	CVBasic	z88dk	ugBASIC	SGlib_CV	PVColLib
Two directions/fire ports	Yes	Yes	`joystick(1/2)`	`JOY(0/1)` backend	NMI snapshots both ports	`joypad_1/2`, four fire bits
Two keypad ports	Yes	Yes	`joystick(3/4)` high byte	`SCANCODE` path; exact port behavior pending	`SG_readCVNumPad`	`keypad_1/2`
Pressed/released	First-class per property	Manual previous state	Manual previous state	Manual previous state	First-class status helpers	Manual previous-state comparison
Spinner/Roller	Yes	Yes	Not found in Coleco target	Not found yet	Not found	`spinner_1/2`, reset and enable APIs
Automatic minimal backend	Yes, capability selected	Runtime-integrated	Linker includes referenced routines	Deployment system includes used modules	Library/NMI input service	Linker extracts referenced modules

Sound and music

Area	Amy Studio	CVBasic	z88dk	ugBASIC	devkitSMS	PVColLib
SN76489 tones/noise	BIOS sound tables and direct formats	`SOUND`	PSG libraries/manual ports	SN76489 backend	PSGlib_CV	BIOS-style sound tables
Sequenced music	BIOS format and Tiny Sound	`MUSIC`, simple/full players	External/player libraries	`MUSIC` backend	PSG streams; looping/status	NMI-serviced music sequences
Concurrent SFX	Coleco table areas/Tiny Sound	Channel depends on music mode	Library-dependent	Pending measurement	PSG SFX channels and frame service	Four music areas plus sound areas 5+ for SFX
Digital samples	DSOUND	No first-class support	Manual/custom	Pending	No first-class support found	No first-class support found
Authoring	Integrated table inspector, tone preview, echo-tail builder, MIDI capture; full sequence editing pending	Note-oriented BASIC source	External VGM/tools	Source/resource conversion	External PSG tools	External table/asset tools

Amy has the broadest playback formats, while CVBasic keeps the simplest music source syntax. Amy Studio can now inspect existing BIOS tables, preview generated commands, audition steady/fade/ echo envelopes, and capture note, velocity, and duration from Web MIDI. The remaining authoring gap is a complete multi-command sequence editor with byte-exact source/project write-back.

Compression evidence

Compression must be scored as **`compressed payload + linked decompressor`**, with destination and cycles. Counting a host compressor without a ColecoVision decoder is invalid.

All fifteen integrated Amy codecs were also compiled as separate Balanced ROMs and run for 180 NTSC frames in GearColeco. Every ROM reproduced Warrior's 6,144 Pattern bytes and 6,144 Color bytes exactly in VRAM. Run **`node tools/test-integrated-codec-vram-roms.mjs`** to repeat this check.

Direct-to-VRAM speed ranking

This benchmark measures two complete 6,144-byte Amy decompression commands (Pattern and Color) with NMI disabled. GearColeco counts exact Z80 master-clock cycles between executable markers and then verifies all 12,288 VRAM bytes. The samples are the mechanically selected most-compressible, median, and least-compressible corpus pictures. Run **`node tools/benchmark-codec-vram-cycles.mjs`**.

Rank	Codec	Decoder bytes	Average cycles	Observed range	NTSC frames	PAL frames
1	MDK-RLE	46	445,166	397,701-480,470	7.45	6.24
2	LZF	117	2,228,378	1,699,652-2,499,555	37.30	31.23
3	Nibble	115	2,500,641	1,109,539-4,605,294	41.86	35.04
4	BitBuster	166	2,534,837	2,200,323-2,848,558	42.43	35.52
5	ZX2	115	2,640,553	2,523,782-2,813,703	44.20	37.00
6	ZX1	127	2,649,233	2,542,465-2,824,788	44.35	37.12
7	ZX0	133	2,776,094	2,593,812-2,917,877	46.47	38.90
8	aPLib Compact	244	2,882,985	2,038,419-3,910,905	48.26	40.40
9	Pletter	212	3,011,986	2,593,248-3,420,132	50.42	42.21
10	ZX7	136	3,045,957	2,646,525-3,421,800	50.99	42.68
11	DAN3	205	3,260,654	2,819,207-3,514,505	54.58	45.69
12	MegaLZ	162	3,545,056	3,003,617-4,081,177	59.35	49.68
13	DAN2	212	3,683,858	2,806,792-4,725,242	61.67	51.62
14	DAN1	205	3,809,391	2,827,513-5,013,133	63.77	53.38
15	Exomizer 2	226	5,443,661	3,379,601-8,078,445	91.13	76.28

Frame values are time equivalents (**`59,736`** cycles NTSC; **`71,364`** PAL), not VBlank waits. Speed and ROM-size rankings are separate: choose using compressed payload plus decoder size, then check whether the measured loading time suits the game. Nibble and aPLib vary substantially by input, which is why Amy Studio should present per-asset estimates rather than one universal winner.

Solution	Confirmed formats	Direct VRAM status	Integrated selection
Amy Studio	Exomizer 2, ZX0, ZX1, ZX2, aPLib, MegaLZ, ZX7, Pletter, DAN1/2/3, LZF, BitBuster, MDK-RLE, Nibble	ColecoVision paths, including workspace-based formats	Browser comparison/import and asset metadata
CVBasic	Pletter	**`DEFINE CHAR/COLOR/SPRITE/VRAM PLETTER`**	Explicit source keyword
z88dk	ZX0/1/2/7 and aPLib families, multiple speed/size decoders	Stock decoders target RAM; ZX0 *and ZX7* Coleco VRAM adaptations verified here	Manual headers/linking and host tools
ugBASIC	MSC1 and RLE types in compiler source	MSC1 image fallback verified; RLE is not implemented for Coleco	Resource compiler can choose compression when it wins

Solution	Confirmed formats	Direct VRAM status	Integrated selection
SGlib_CV / SMSlib routine	ZX7 and aPLib	ZX7 API; aPLib direct-to-VRAM exact with frame IRQ disabled	Manual host compression and C asset inclusion
PVColLib	RLE, Pletter, DAN1/2/3	RLE direct-to-VRAM verified exactly here; other paths remain separate candidates	`gfx2col` conversion and C asset inclusion
NewColeco	GETPUT 1.1 MDK-RLE	Direct-to-VRAM verified exactly on Warrior	External CVPaint/asset tools and C data inclusion

z88dk's public ZX0 integration request dates from February 2021, consistent with ZX0/ZX7 being available there by spring 2021. Its bundled compressor identifies itself as ZX0 v1.5 and emits the official classic v1 format. Amy Studio's current **`zx0`** codec emits the later official v2 format by default. The streams are intentionally not interchangeable. z88dk's stock routines target RAM; this benchmark adds only ColecoVision TMS9918 direct-to-VRAM adaptations and validates their output independently.

Codec naming and integration policy

Candidate	Honest Amy name	Current evidence	Integration condition
ZX0	**`ZX0`** / codec **`zx0`**	Existing v2 browser encoder and Coleco VRAM decoder	Keep as the default
ZX1	**`ZX1`** / codec **`zx1`**	Byte-identical browser encoder; exact GearColeco VRAM and cycle proof	Integrated and measured
ZX2	**`ZX2`** / codec **`zx2`**	Exact corpus round-trip, five-profile VRAM proof, and cycle proof	Integrated and measured
aPLib	**`aPLib`** / codec **`aplib`**	Bidirectional appack parity and exact Amy/GearColeco VRAM ROM	Integrated; keep NMI-safe upload and attribution explicit
MegaLZ	**`MegaLZ`** / codec **`megalz`**	Independent DEC40-compatible encoder; exact corpus round-trip and GearColeco VRAM/cycle proof	Integrated and measured
Exomizer 2	codec **`exomizer`**	Independent browser encoder, exact 42-picture round trip, five-profile VRAM proof, and cycle proof	Integrated; 226-byte decoder; compiler-assigned 256-byte-aligned **`AMY_EXOMIZER_TABLE`** reserves 156 bytes through address +$009B
MSC1	**`MSC1`**	ugBASIC discards it for Warrior when it gives no gain	Useful Coleco corpus wins and a VRAM strategy

A host compressor alone is insufficient. An Amy codec requires round-trip tests, exact GearColeco VRAM output, decoder cost, explicit destination semantics, malformed-stream failure tests, documentation, and attribution. RAM-only APIs and locally written VRAM ports must remain labelled.

Current verdict: Amy leads in codec breadth and ColecoVision workflow; devkitSMS has verified ZX7 and aPLib paths, PVColLib has a verified RLE path, and CVBasic has a concise Pletter path. On Warrior and Cake, official ZX1, ZX2, and aPLib do not beat Amy's ZX0 first-use ROM size. aPLib nevertheless cuts the devkitSMS Warrior ROM from 13,680 raw bytes to 4,713 occupied bytes. ZX1 is now an integrated Amy codec with a 127-byte direct-to-VRAM decoder and exact runtime and cycle proofs. ugBASIC's MSC1 is a valid comparison candidate, not ten separate codecs, but it does not improve the Warrior resource.

What MSC1 actually is

MSC1 is ugBASIC's compact block-repetition format, not another general ZX0-style LZ codec. A stream contains literal blocks of 1-127 bytes and back-references that repeat one four-byte sequence up to 32 times. The back-reference offset is limited to roughly 1 KB, and a zero token ends the stream. This favors maps, records, planar graphics, and other data with repeated groups of four bytes.

The current Z80 decoder is simple and Apache-2.0 licensed, but writes to ordinary memory with **`LD (DE),A`**. On a stock 1 KB ColecoVision it cannot directly expand a 6 KB pattern or color table. More importantly, ugBASIC's generic **`LOAD("file") COMPRESSED`** implementation only invokes MSC1 when the target declares expansion banks. The standard Coleco target does not, so the keyword is accepted but ignored there. Five raw/compressed resource pairs produced byte-identical generated code and data. This is therefore not presently a

general Coleco capability comparable to Amy's direct-to-VRAM codecs.

MSC1 can still be selected by some ugBASIC image/resource conversion paths. A fair evaluation of that narrower feature needs a direct-to-VRAM proof and a mixed corpus: bitmap tables, name tables, tile sets, sprite frames, level maps, record arrays, and sound data. Only candidates that reduce **`payload + linked decoder`**, round-trip exactly, and pass GearColeco should be added.

aPLib integration status

aPLib has three useful pieces already available locally: the official host compressor, Amy's JavaScript beam-search encoder, and direct-to-VRAM Z80 implementations in z88dk/devkitSMS. The devkitSMS routine uses the Coleco-compatible **`$BF/$BE`** VDP ports, but is explicitly unsafe while the display is active and lacks the Coleco BIOS protection used by **`SG_decompressZX7toVRAM`**.

The JavaScript encoder now passes bidirectional differential checks against official aPLib on real TMS9918 tables and structured, incompressible, repetitive, short, and boundary-sized data. Its exact-cost planner produces 3,054 bytes for Cake and 2,973 for Warrior, versus official raw payloads of 3,088 and 2,975. Official **`appack`** still wins some individual resources, so neither encoder universally dominates.

Amy now ships the browser encoder and a 244-byte compact public-domain SMSlib-derived ColecoVision VRAM decoder. **`decompress aplib Source to vram.pattern`** accepts a raw **`.aplib`** stream and the existing VRAM upload wrapper provides the same NMI-safe contract as other Amy LZ codecs. A compiled Amy ROM reconstructed Warrior's 6,144-byte pattern and color tables exactly in GearColeco under all five optimization profiles. The Balanced test ROM occupied 3,517 bytes with 2,973 compressed data bytes. The official aPLib executable is used only for differential testing and is not redistributed.

The first reproducible size results and the rules for the cross-tool runtime comparison are in **`competition/benchmarks/compression/README.md`**. On two real 12,288-byte TMS9918 pictures, ZX0 has the smallest Amy first-use ROM total after its current 133-byte decompressor estimate is included. The separate GearColeco cycle benchmark supplies the speed evidence; size and speed remain independent selection criteria.

Memory, ROM size, and banking

Area	Amy Studio	CVBasic	z88dk	ugBASIC	devkitSMS	PVColLib
Stock 1 KB RAM focus	Core design, RAM estimates, overlays	Yes, global/static model	Configurable CRT/C runtime	Backend manages runtime/resources	SDCC/static library model	SDCC/static library model
Dead helper elimination	Capability-driven generation	Compiler-generated runtime	Linker sections/libraries	Deploy-on-use modules and target optimizer	Linker library extraction	Linker library extraction
Optimizer	Five profiles plus runtime corpus	Z80 optimizer and peepholes	sccz80/zsdcc optimizers	Coleco-specific optimizer source	SDCC optimizer/peepholes	SDCC size optimization
Beyond 32 KB	Deliberately excluded	MegaCart up to 1 MB	Coleco banking/toolchain	Target support not yet proven	MegaCart and banked functions documented	MegaCart tools and example verified
Debug-aware RAM names	Yes, including overlay aliases	Assembly labels	Map/debug symbols	Generated symbols	Map symbols	Map symbols

Banking is a capability advantage for CVBasic, z88dk, and devkitSMS, but not a feature Amy intends to adopt: Amy explicitly preserves the original unexpanded-hardware philosophy. The comparison should describe this limitation honestly without turning it into a roadmap requirement.

Stock baseline versus expanded hardware

The size and runtime ranking above targets an original ColecoVision: TMS9918A video, SN76489 sound, 1 KB system RAM, Coleco BIOS, and an unbanked cartridge whenever possible. Optional hardware is credited separately because it changes the machine available to the programmer:

Extension	Verified or documented advantage	Scope in this comparison
MegaCart / bank switching	CVBasic, z88dk, devkitSMS, and PVColLib can exceed the normal cartridge space	Valid capability; excluded from stock-ROM size ranking
F18A	PVColLib provides dedicated APIs and examples for enhanced video hardware	Valid for modified or compatible clone systems; out of scope for ColecoVision-exclusive titles
SGM / AY-3-8910-compatible sound	CVBasic and PVColLib provide explicit SGM paths; PVColLib also detects SGM RAM and ADAM	Valid expansion/clone capability; out of scope for the stock sound ranking
Extra RAM	Available through SGM, ADAM, and compatible clones depending on the tool/runtime	Report separately; never count it as stock 1 KB RAM

This separation lets every solution show its extended-hardware strengths without weakening Amy Studio's deliberate original-hardware target.

Development experience

Area	Amy Studio	CVBasic	z88dk	ugBASIC	devkitSMS	PVColLib
Browser IDE	Integrated	No official integrated IDE	No	IDE/web options advertised	No	No
Source breakpoints	Yes	External emulator	External debugger	IDE-dependent/unverified	External emulator	External emulator
ASM stepping/rewind	Integrated GearColeco workflow	External	External	External	External	External
RAM/VRAM/symbol inspection	Integrated	External	External	External	External	External/map symbols
Cycle profiling	Integrated	External/manual	External/manual	Unverified	External/manual	External/manual
Project graphics editors	Integrated/configurable	Separate tools	Separate tools	Resource conversion/IDE	Separate tools	`gfx2col`/separate tools
Automated ROM tests	Corpus, checkpoints, runtime harness	Not integrated	Buildable manually	Not established	Not established	Not established

This is Amy Studio's decisive advantage. A fair comparison should still label command-line-only Amy scripts separately from features directly accessible in the IDE.

Amy Studio leads in ColecoVision integration

Amy's strongest distinction is not one isolated keyword. It connects source editing, asset conversion, compression, assembly optimization, ROM execution, source breakpoints, rewind, memory and VRAM inspection, controller configuration, profiling, and automated tests in one ColecoVision-focused workflow. Its overlays, typed state machines, BCD, fixed-point support, runtime-checked wide integers, collision helpers, and BIOS-aware input selection are also substantial language-level strengths.

CVBasic leads in portability and established BASIC simplicity

CVBasic has a compact QBasic-like surface, a mature ColecoVision backend, many complete game examples, spinner support, sprite flicker support, music commands, and optional ROM banking. It also targets many related machines. It does not offer Amy's records, overlays, wide numeric model, integrated debugger, or asset/debug pipeline.

Official source: [nanochess/CVBasic](https://github.com/nanochess/CVBasic)

z88dk leads in general C and toolchain breadth

z88dk provides full C data structures, pointers, mature compilers, assemblers, linkers, libraries, compression utilities, and many Z80 targets. This flexibility also exposes more low-level choices to the programmer. Its generic graphics or library facilities must not be mistaken for an Amy-style ColecoVision game API without target-specific proof.

Official source: [z88dk/z88dk](https://github.com/z88dk/z88dk)

ugBASIC has the most ambitious portable high-level vocabulary

ugBASIC documents nonblocking animation, movement, paths, image conversion, sprites, sound, and many modern BASIC constructs. This makes it especially relevant when considering future Amy animation ergonomics. However, its multi-target manual describes capabilities that can differ by backend. Generated ColecoVision Z80 and runtime behavior must be measured before declaring those features equivalent.

Official sources: [ugBASIC site](https://ugbasic.iwashere.eu/) and [spotlessmind1975/ugbasic](https://github.com/spotlessmind1975/ugbasic)

devkitSMS offers a compact C game-library workflow

devkitSMS combines SDCC with small game-oriented libraries and asset tools. **`SGlib_CV`** and **`PSGlib_CV`** make it relevant to ColecoVision even though much of the surrounding documentation is SMS-oriented. Only APIs present in the CV headers and libraries should enter the final score.

The local `SGlib_CV` source confirms that its ColecoVision NMI scans direction/fire mode and numpad/right-trigger mode, preserves current and previous states, and implements `SG_getKeysStatus`, `SG_getKeysPressed`, `SG_getKeysHeld`, `SG_getKeysReleased`, and `SG_readCVNumPad`. These input capabilities are source-verified for ColecoVision, not inferred from SMSlib.

Official source: [sverx/devkitSMS](https://github.com/sverx/devkitSMS)

PVColLib offers broad ColecoVision-focused C APIs

PVColLib is a ColecoVision-focused SDCC library and development kit. Its official repository includes the compiler toolchain, VDP and controller APIs, sprites, sound/music support, compression routines, MegaCart, SGM, Phoenix, and F18A facilities. The three stock fixtures here use its official build flags and linked library. Local headers and examples verify both controller ports and keypads, both spinners, 32 sprites, NMI sprite swapping, hitbox collisions, BIOS-style sound tables, sequenced music with concurrent SFX areas, direct VRAM operations, MegaCart, SGM, Phoenix, and F18A support. Its RLE bitmap path was validated byte-for-byte in VRAM; the available Pletter and DAN paths are not credited with a benchmark size until a compatible stream passes the same exact runtime test.

Official source: [alekmaul/pvcollib](https://github.com/alekmaul/pvcollib)

NewColeco remains a strong historical baseline

The recovered SDCC branch uses Amy's original `CRTCV`, `CVLIB`, and GETPUT 1.1 libraries. Its prebuilt ASxxxx objects link successfully with the locally available SDCC 3.8 toolchain after using the modern `.rel` extension; no ABI or library source change was required. The three new fixtures complete 180 NTSC frames in GearColeco. Warrior is decompressed directly to VRAM with GETPUT's MDK-RLE routine and matches all 12,288 target table bytes and all 49,152 pixels.

This is not an unrelated seventh competitor: it is the documented historical ancestor of Amy Studio. Its 795-byte Hello, 3,643-byte DAN2 bitmap, and 932-byte controller monitor show that the old BIOS-aware and link-only-what-is-used philosophy was already effective. Amy Studio improves those results while adding the Amy language, integrated assets, diagnostics, and debugging.

Next measurement suite

The comparison becomes stronger as equivalent game behaviors are implemented seven times. Current status and order:

1. **Controller snapshot: complete.** Six toolchains pass injected neutral, keypad, direction,

fire, and release states; ugBASIC's VDP R7 update remains the documented exception.

2. **Sprite/metasprite stress: complete.** Seven toolchains build and boot the shared fixture.

Amy's native metasprite path is runtime-equivalent, uses zero permanent RAM, preserves explicit layer priority, and is smaller than the manual renderer in all five profiles.

3. **Tile animation: complete.** Seven ROMs animate shared patterns and 3x2 Name Table frames;

GearColeco verifies every pattern, color, tile position, animation phase, and final state.

4. **State update: complete.** Seven ROMs run the same actor array, collision checks, timers, and

state dispatch oracle.

5. **Sound authoring: functional and still evolving.** BIOS commands and Tiny Sound sequences can

be inspected, auditioned, edited, imported, and written back; UX and audio-parity QA continue.

6. **Compression: payload and cycle suites complete.** Fifteen codecs have exact VRAM proof;

representative streams now include linked decoder bytes and GearColeco cycle measurements.

7. **Visible Hello: complete.** Keep it separate from the minimal runtime fixture so font/text

costs remain explicit.

All six Amy listings are available in Amy Studio under **Benchmarks**. The evidence archive groups the matching source and loadable ROM for every tool. Measurements use occupied bytes rather than padding, and a smaller result counts only when the shared runtime oracle passes.

Ranked Amy gap plan

Rank	Work	Value	Effort	Risk	Decision gate
1	Isolate and reduce display code cost	High	Medium	Low	Measure screen setup, literal text, coordinates, decimal conversion, and VRAM output separately
2	Close sound-editor fidelity and UX gaps	High	Medium	Medium	Emulator-faithful preview, Tiny envelopes, reliable edits, undo, and byte-exact project insertion
3	Measure tile-animation cycles and VRAM budget	Medium	Small	Low	Runtime equivalence and corruption oracle pass; add comparable cycle/write counts
4	Finish in-Studio compression guidance	Medium	Small	Low	Show first-use ROM, CPU RAM, and measured cycles beside each asset choice
5	Small explicit animation service	High	Large	Medium-high	Add only after benchmark evidence; zero linked cost when unused and visible RAM/cycle budget
6	Nested aggregate 2D fields and final operand symmetry	Medium	Medium	Medium	Direct record/overlay 2D fields already pass; add nesting only for a real game need

Completed gaps and next concrete work

Completed and runtime-guarded:

- seven-tool metasprite fixture, SAT oracle, native metasprites, protected-priority flicker;
- primitive 2D record/overlay fields, verified in all five profiles;
- fifteen direct-to-VRAM codecs with exact round trips, sizes, cycles, and 42-picture rankings;
- exact Warrior proof ROMs: Exomizer 3,319 bytes and ZX0 3,254 bytes;
- seven-tool state-update fixture: all engines reach the same oracle. Amy is 1,460 bytes displayed

and 949 bytes engine-only in Experimental; compound record-array expressions now pass. Default numeric output shed 68 ROM bytes and two RAM bytes by linking glyph remapping only when used;

- seven-tool tile animation: all engines animate the same shared star patterns and six 3x2 ships;

exact VRAM checks also exposed and fixed indexed-coordinate `put frame` source corruption;

- sound inspection, continuous playback, sequencer editing, undo, import, and Web MIDI.

Next concrete work is **display-cost isolation**. Measure literal text, coordinate setup, decimal conversion, and VRAM output independently, then optimize only behavior-identical paths. Tile animation runtime equivalence is complete; cycle/write accounting and sound-editor timing QA follow.